data

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2024-03-08

```r
library(stargazer)
```

```
##
## Please cite as:

##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.

##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```r
library(reshape2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)
```

```
##
## Attaching package: 'tidyr'

## The following object is masked from 'package:reshape2':
##
##     smiths
```

```r
library(lmtest)
```

```
## Loading required package: zoo

##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
library(sandwich)
library(knitr)
library(broom)
```

```r
library(readr)
DATA_NL <- read_csv("DATA_NL /DATA_NL.csv")
```

```
## Rows: 8984 Columns: 47
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (47): R0000100, R0536300, R1482600, R9705300, S2001000, S2011301, S31346...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Dictionary-like list mapping years to their respective column names
years_mapping <- list(
  `2003` = list(age = 'S2001000', education = 'S2011301', income = 'S3134600'),
  `2004` = list(age = 'S3801100', education = 'S3812201', income = 'S4799600'),
  `2005` = list(age = 'S5401000', education = 'S5412600', income = 'S6501000'),
  `2006` = list(age = 'S7501200', education = 'S7513500', income = 'S8496500'),
  `2007` = list(age = 'T0008500', education = 'T0013900', income = 'T0889800'),
  `2008` = list(age = 'T2011100', education = 'T2016000', income = 'T3003000'),
  `2009` = list(age = 'T3601500', education = 'T3606300', income = 'T4406000'),
  `2010` = list(age = 'T5201400', education = 'T5206700', income = 'T6055500'),
  `2011` = list(age = 'T6651300', education = 'T6656500', income = 'T7545600'),
  `2013` = list(age = 'T8123600', education = 'T8128900', income = 'T8976700'),
  `2015` = list(age = 'U0001800', education = 'U0008800', income = 'U0956900'),
  `2017` = list(age = 'U1838500', education = 'U1845400', income = 'U2857200'),
  `2019` = list(age = 'U3437900', education = 'U3443900', income = 'U4282300'),
  `2021` = list(age = 'U4943000', education = 'U4949600', income = 'U5753500')
)


# Initialize an empty dataframe to store the combined results
combined_data <- data.frame()

# Function to reshape data for a given year
reshape_year <- function(year_data, year, DATA_NL) {
  reshaped_data <- DATA_NL %>%
    select(Pubid = R0000100, Sex = R0536300, Afqt = R9705300, Annual_hours = Z9065800,
           Age = !!sym(year_data$age),
           Education = !!sym(year_data$education),
           Income = !!sym(year_data$income)) %>%
    mutate(Year = as.integer(year)) %>%
    filter(Age > 0) # Filter out invalid ages

  return(reshaped_data)
}
```

```r
# Iterating over each year and combining the data
for (year in names(years_mapping)) {
  if (!is.null(years_mapping[[year]]$age)) {
    year_data <- reshape_year(years_mapping[[year]], year, DATA_NL)
    combined_data <- rbind(combined_data, year_data)
  }
}
```

Replace negative values with NA

```r
combined_data[combined_data == -4 | combined_data == -5] <- NA
```

```r
combined_data <- combined_data %>% filter(!is.na(combined_data$Annual_hours) & combined_data$Annual_hou
```

```r
# Convert annual income to weekly wages
combined_data$weekly_wages <- combined_data$Income / 52

# Convert weekly wages to hourly wages
combined_data$hourly_wages <- combined_data$weekly_wages  / (combined_data$Annual_hours / 52)

# Filter based on hourly wages
cleaned_data <- combined_data %>% filter(hourly_wages >= 5 & hourly_wages <= 100)

cleaned_data
```

```
## # A tibble: 41,630 x 10
##    Pubid   Sex  Afqt Annual_hours   Age Education Income  Year weekly_wages
##    <dbl> <dbl> <dbl>        <dbl> <dbl>     <dbl>  <dbl> <int>        <dbl>
##  1     1     2     4         1090    22        16  15000  2003        288.
##  2     4     2    NA         2022    22        12  23000  2003        442.
##  3     8     2    NA          312    22        16   5000  2003         96.2
##  4    12     1    NA         1330    22        13  34000  2003        654.
##  5    31     1   607          370    21        15   3000  2003         57.7
##  6    39     1    NA          670    23        16  12000  2003        231.
##  7    42     1   322          300    20        13   5000  2003         96.2
##  8    43     2    NA          560    22        13   5000  2003         96.2
##  9    48     2    NA         1060    20        14  16800  2003        323.
## 10    49     2   190          526    20        12  15000  2003        288.
## # i 41,620 more rows
## # i 1 more variable: hourly_wages <dbl>
```

#log hourly wages

```r
combined_data<- subset(combined_data, hourly_wages > 0)

combined_data$log_hourly_wages <- log(combined_data$hourly_wages)
```

```r
combined_data <- na.omit(combined_data, cols = c("Education", "AFQT", "log_hourly_wages "))
```

```r
# Ensure the independent variables are numeric
combined_data$education <- as.numeric(combined_data$Education)
combined_data$afqt <- as.numeric(combined_data$Afqt)
```

```r
combined_data
```

```
## # A tibble: 19,321 x 13
##    Pubid   Sex  Afqt Annual_hours   Age Education Income  Year weekly_wages
##    <dbl> <dbl> <dbl>        <dbl> <dbl>     <dbl>  <dbl> <int>        <dbl>
##  1     1     2     4         1090    22        16  15000  2003        288.
##  2    31     1   607          370    21        15   3000  2003         57.7
##  3    41     2   405          790    20        14   1000  2003         19.2
##  4    42     1   322          300    20        13   5000  2003         96.2
##  5    49     2   190          526    20        12  15000  2003        288.
##  6    56     2  1024           40    21        15   9000  2003        173.
##  7    62     1   225           60    23        16   3000  2003         57.7
##  8   105     1   305          240    20        12  23000  2003        442.
##  9   111     2   467         1200    22        16   6000  2003        115.
## 10   113     1    55          360    19        12   1200  2003         23.1
## # i 19,311 more rows
## # i 4 more variables: hourly_wages <dbl>, log_hourly_wages <dbl>,
## #   education <dbl>, afqt <dbl>
```

```r
summary_statistics <- combined_data %>%
  summarise(
    Mean_Log_Wages = mean(log_hourly_wages, na.rm = TRUE),
    SD_Log_Wages = sd(log_hourly_wages, na.rm = TRUE),
    N_Log_Wages = sum(!is.na(log_hourly_wages)),

    Mean_Education = mean(Education, na.rm = TRUE),
    SD_Education = sd(Education, na.rm = TRUE),
    N_Education = sum(!is.na(Education)),

    Mean_AFQT = mean(Afqt, na.rm = TRUE),
    SD_AFQT = sd(Afqt, na.rm = TRUE),
    N_AFQT = sum(!is.na(Afqt)),

    Mean_Age = mean(Age, na.rm = TRUE),
    SD_Age = sd(Age, na.rm = TRUE),
    N_Age = sum(!is.na(Age)),

    Mean_Annualh = mean(Annual_hours, na.rm = TRUE),
    SD_Anuualh = sd(Annual_hours, na.rm = TRUE),
    N_Anualh = sum(!is.na(Annual_hours)),

    Mean_Income = mean(Income, na.rm = TRUE),
    SD_Income = sd(Income, na.rm = TRUE),
    N_Income = sum(!is.na(Income)),
  )
```

```r
# To view the result
print(summary_statistics)
```

```
## # A tibble: 1 x 18
##   Mean_Log_Wages SD_Log_Wages N_Log_Wages Mean_Education SD_Education
##            <dbl>        <dbl>       <int>          <dbl>        <dbl>
## 1           3.66         1.43       19321           15.1         3.33
## # i 13 more variables: N_Education <int>, Mean_AFQT <dbl>, SD_AFQT <dbl>,
## #   N_AFQT <int>, Mean_Age <dbl>, SD_Age <dbl>, N_Age <int>,
## #   Mean_Annualh <dbl>, SD_Anuualh <dbl>, N_Anualh <int>, Mean_Income <dbl>,
## #   SD_Income <dbl>, N_Income <int>
```

```r
library(dplyr)


summary_statistics <- combined_data %>%
  group_by(Sex) %>%
  summarise(
    Mean_Log_Wages = mean(log_hourly_wages, na.rm = TRUE),
    SD_Log_Wages = sd(log_hourly_wages, na.rm = TRUE),
    N_Log_Wages = sum(!is.na(log_hourly_wages)),

    Mean_Education = mean(Education, na.rm = TRUE),
    SD_Education = sd(Education, na.rm = TRUE),
    N_Education = sum(!is.na(Education)),

    Mean_AFQT = mean(Afqt, na.rm = TRUE),
    SD_AFQT = sd(Afqt, na.rm = TRUE),
    N_AFQT = sum(!is.na(Afqt)),

    Mean_Age = mean(Age, na.rm = TRUE),
    SD_Age = sd(Age, na.rm = TRUE),
    N_Age = sum(!is.na(Age)),

    Mean_Annualh = mean(Annual_hours, na.rm = TRUE),
    SD_Anuualh = sd(Annual_hours, na.rm = TRUE),
    N_Anualh = sum(!is.na(Annual_hours)),

    Mean_Income = mean(Income, na.rm = TRUE),
    SD_Income = sd(Income, na.rm = TRUE),
    N_Income = sum(!is.na(Income)),
  )

# To view the result
print(summary_statistics)
```

```
## # A tibble: 2 x 19
##     Sex Mean_Log_Wages SD_Log_Wages N_Log_Wages Mean_Education SD_Education
##   <dbl>          <dbl>        <dbl>       <int>          <dbl>        <dbl>
## 1     1           3.73         1.45       10419           14.8         2.95
## 2     2           3.58         1.39        8902           15.6         3.68
## # i 13 more variables: N_Education <int>, Mean_AFQT <dbl>, SD_AFQT <dbl>,
```

```
## #   N_AFQT <int>, Mean_Age <dbl>, SD_Age <dbl>, N_Age <int>,
## #   Mean_Annualh <dbl>, SD_Anuualh <dbl>, N_Anualh <int>, Mean_Income <dbl>,
## #   SD_Income <dbl>, N_Income <int>
```

```
men_only <- filter(combined_data, Sex == 1)
women_only <- filter(combined_data, Sex == 2)

men_edu_less_16 <- filter(combined_data, Sex == 1 & Education < 16)
men_edu_16_or_more <- filter(combined_data,Sex == 1 & Education >= 16)
women_edu_less_16 <- filter(combined_data, Sex == 2 & Education < 16)
women_edu_16_or_more <- filter(combined_data, Sex == 2 & Education >= 16)


run_regression <- function(combined_data) {
  model <- lm(log_hourly_wages ~ Education+ Afqt, data = combined_data)
  return(summary(model))
}
```

Running regression on each subsample and print the results

```
cat("Men Only:\n")
```

```
## Men Only:
```

```
print(run_regression(men_only))
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = combined_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.9152  -0.8069  -0.0782   0.7633   5.7243
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.677e+00  6.929e-02  24.209  < 2e-16 ***
## Education   1.302e-01  4.815e-03  27.032  < 2e-16 ***
## Afqt        2.053e-04  2.838e-05   7.234 5.02e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.386 on 10416 degrees of freedom
## Multiple R-squared:  0.08639,    Adjusted R-squared:  0.08621
## F-statistic: 492.5 on 2 and 10416 DF,  p-value: < 2.2e-16
```

```
cat("\nWomen Only:\n")
```

```
##
## Women Only:
```

```r
print(run_regression(women_only))
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = combined_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.4075 -0.7807 -0.0110  0.8090  5.0590
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 2.024e+00  6.256e-02  32.351  < 2e-16 ***
## Education   9.568e-02  3.915e-03  24.441  < 2e-16 ***
## Afqt        1.124e-04  3.256e-05   3.452 0.000559 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.344 on 8899 degrees of freedom
## Multiple R-squared:  0.06802,    Adjusted R-squared:  0.06781
## F-statistic: 324.7 on 2 and 8899 DF,  p-value: < 2.2e-16
```

#Men with Education <16

```r
cat("\nMen with Education < 16:\n")
```

```
##
## Men with Education < 16:
```

```r
print(run_regression(men_edu_less_16))
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = combined_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.7876 -0.7189 -0.1171  0.6204  5.6229
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.240e+00  1.312e-01  17.075  < 2e-16 ***
## Education    8.248e-02  1.031e-02   7.998 1.53e-15 ***
## Afqt        -1.028e-05  4.296e-05  -0.239    0.811
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.297 on 5607 degrees of freedom
## Multiple R-squared:  0.01146,    Adjusted R-squared:  0.0111
## F-statistic: 32.49 on 2 and 5607 DF,  p-value: 9.327e-15
```

```r
cat("\nMen with Education > 16:\n")
```

```
##
## Men with Education > 16:
```

```r
print(run_regression(men_edu_16_or_more))
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = combined_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.3006 -0.8730  0.0004  0.8161  5.4373
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.649e+00  1.541e-01  23.682  < 2e-16 ***
## Education   2.441e-02  8.902e-03   2.742  0.00612 **
## Afqt        2.196e-04  3.808e-05   5.767 8.57e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.436 on 4806 degrees of freedom
## Multiple R-squared:  0.008994,   Adjusted R-squared:  0.008582
## F-statistic: 21.81 on 2 and 4806 DF,  p-value: 3.725e-10
```

```r
cat("\nWomen with Education < 16:\n")
```

```
##
## Women with Education < 16:
```

```r
print(run_regression(women_edu_less_16))
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = combined_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.0977 -0.6940 -0.0431  0.7603  4.9372
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 2.579e+00  1.274e-01  20.246  < 2e-16 ***
## Education   3.218e-02  9.652e-03   3.334 0.000865 ***
## Afqt        2.920e-05  5.301e-05   0.551 0.581849
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.262 on 3680 degrees of freedom
## Multiple R-squared:  0.003265,   Adjusted R-squared:  0.002723
## F-statistic: 6.027 on 2 and 3680 DF,  p-value: 0.002438
```

****

```r
cat("\nWomen with Education > 16:\n")
```

```
##
## Women with Education > 16:
```

```r
print(run_regression(women_edu_16_or_more))
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = combined_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.8682 -0.7586  0.0090  0.7687  5.0407
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.330e+00  9.597e-02  34.700  < 2e-16 ***
## Education   3.697e-02  5.269e-03   7.016 2.57e-12 ***
## Afqt        4.541e-06  4.020e-05   0.113     0.91
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.333 on 5216 degrees of freedom
## Multiple R-squared:  0.009364,   Adjusted R-squared:  0.008984
## F-statistic: 24.65 on 2 and 5216 DF,  p-value: 2.21e-11
```

afqt not significant

Men Only Regression:

Education has a significant positive effect on log hourly wages, suggesting higher returns for each additional year of schooling. AFQT scores also positively affect log hourly wages and are significant, indicating that cognitive skills are valued in the labor market for men. Women Only Regression:

Education, similar to the men's model, has a positive and significant effect on log hourly wages for women. The coefficient for AFQT is positive and significant, implying that for women, like men, cognitive skills contribute positively to their earnings. Men with Education < 16:

Education remains a significant factor in predicting wages, but the coefficient is slightly lower than in the general male population, suggesting possibly lower returns for those with education below the standard college level. AFQT score does not have a significant impact on wages in this subset, which could suggest that for men with lower education levels, other factors might play a more critical role in wage determination. Men with Education > 16:

Higher education continues to be positively correlated with wages, with a significant coefficient indicating that men with education beyond the 16-year mark experience higher wage returns. AFQT scores show a significant positive effect, aligning with the notion that cognitive skills paired with higher education level are valuable. Women with Education < 16:

The impact of education on wages is significant and positive, but the magnitude of the coefficient is lower compared to women in the general population, hinting at lower wage returns for lower levels of education. AFQT scores do not significantly impact wages for this group, suggesting that for women with less than 16

years of education, cognitive ability may not be as strongly linked to wages as other variables. Women with Education > 16:

The coefficient for education is significant, indicating that women with more than 16 years of education see considerable wage benefits. The AFQT coefficient is not significant, which could indicate that for highly educated women, the labor market rewards education itself rather than test-measured cognitive skills.2

```r
# Function to run regression for a given age range
run_regression <- function(combined_data, age_min, age_max) {
  subset_data <- combined_data %>%
    filter(Age >= age_min & Age <= age_max)

  result <- lm(log_hourly_wages ~ Education + Afqt, data = subset_data)
  return(summary(result))
}

# Run regressions for different age groups
regression_20_25 <- run_regression(combined_data, 20, 25)
regression_25_30 <- run_regression(combined_data, 25, 30)
regression_30_35 <- run_regression(combined_data, 30, 35)
```

```r
print("Regression for Ages 20-25")
```

```
## [1] "Regression for Ages 20-25"
```

```r
print(regression_20_25)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = subset_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.0616 -0.7317 -0.0918  0.7144  5.4306
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.812e+00  1.051e-01  17.247   <2e-16 ***
## Education   8.464e-02  7.473e-03  11.326   <2e-16 ***
## Afqt        9.776e-06  3.540e-05   0.276    0.782
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.261 on 6259 degrees of freedom
## Multiple R-squared:  0.022,  Adjusted R-squared:  0.02169
## F-statistic: 70.41 on 2 and 6259 DF,  p-value: < 2.2e-16
```

```r
print("Regression for Ages 25-30")
```

```
## [1] "Regression for Ages 25-30"
```

```r
print(regression_25_30)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = subset_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.5410  -0.7050  -0.0601   0.6858   5.8362
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.739e+00  7.780e-02  22.349  < 2e-16 ***
## Education   1.185e-01  5.224e-03  22.689  < 2e-16 ***
## Afqt        2.300e-04  3.088e-05   7.449 1.04e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.255 on 7688 degrees of freedom
## Multiple R-squared:  0.08367,    Adjusted R-squared:  0.08343
## F-statistic:   351 on 2 and 7688 DF,  p-value: < 2.2e-16
```

```r
print("Regression for Ages 30-35")
```

```
## [1] "Regression for Ages 30-35"
```

```r
print(regression_30_35)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = subset_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.5074 -0.7829 -0.0558  0.7624  5.1615
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 2.9362606  0.0782778  37.511  < 2e-16 ***
## Education   0.0592413  0.0047681  12.425  < 2e-16 ***
## Afqt        0.0003229  0.0000439   7.356 2.25e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.375 on 4344 degrees of freedom
## Multiple R-squared:  0.05329,    Adjusted R-squared:  0.05285
## F-statistic: 122.3 on 2 and 4344 DF,  p-value: < 2.2e-16
```

```r
data_young <- subset(combined_data, Age <= 30)

data_old <- subset(combined_data, Age > 30)
```

```r
# Younger group, men only

model_young_men <- lm(log_hourly_wages ~ Education + Afqt, data = data_young, subset = Sex == 1)
summary(model_young_men)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = data_young,
##     subset = Sex == 1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.0309  -0.7284  -0.0998   0.7081   6.0206
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.6387866  0.0878954   18.64  < 2e-16 ***
## Education   0.1157093  0.0062825   18.42  < 2e-16 ***
## Afqt        0.0001530  0.0000337    4.54 5.71e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.304 on 6814 degrees of freedom
## Multiple R-squared:  0.0632, Adjusted R-squared:  0.06292
## F-statistic: 229.8 on 2 and 6814 DF,  p-value: < 2.2e-16
```

```r
# older group, men only
model_older_men <- lm(log_hourly_wages ~ Education + Afqt, data = data_old, subset = Sex == 1)
summary(model_older_men)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = data_old,
##     subset = Sex == 1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.4049  -0.7876  -0.1184   0.7103   4.8519
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 2.508e+00  1.066e-01  23.529  < 2e-16 ***
## Education   1.046e-01  7.049e-03  14.843  < 2e-16 ***
## Afqt        3.461e-04  4.647e-05   7.446 1.19e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.377 on 3599 degrees of freedom
## Multiple R-squared:  0.09113,    Adjusted R-squared:  0.09063
## F-statistic: 180.4 on 2 and 3599 DF,  p-value: < 2.2e-16
```

```r
# Younger group, women only
model_young_women <- lm(log_hourly_wages ~ Education + Afqt, data = data_young, subset = Sex == 2)
summary(model_young_women)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = data_young,
##     subset = Sex == 2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.1948 -0.7245 -0.0094  0.7888  4.4183
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.0183667  0.1024703   9.938   <2e-16 ***
## Education   0.1507484  0.0068337  22.060   <2e-16 ***
## Afqt        0.0000160  0.0000387   0.413    0.679
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.272 on 5791 degrees of freedom
## Multiple R-squared:  0.08163,    Adjusted R-squared:  0.08131
## F-statistic: 257.4 on 2 and 5791 DF,  p-value: < 2.2e-16
```

```r
# older group,women only
model_older_women <- lm(log_hourly_wages ~ Education + Afqt, data = data_old, subset = Sex == 2)
summary(model_older_women)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = data_old,
##     subset = Sex == 2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.4307 -0.7531  0.0120  0.7514  4.8056
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.102e+00  8.355e-02  37.124  < 2e-16 ***
## Education   5.119e-02  4.748e-03  10.781  < 2e-16 ***
## Afqt        2.421e-04  5.479e-05   4.419 1.03e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.342 on 3105 degrees of freedom
## Multiple R-squared:  0.04575,    Adjusted R-squared:  0.04513
## F-statistic: 74.43 on 2 and 3105 DF,  p-value: < 2.2e-16
```

```r
# Calculate the experience
combined_data$Experience <- combined_data$Age - combined_data$Education - 6
```

```r
# Function to run regression for a given experience range
run_regression_experience <- function(DATA_NL, exp_min, exp_max) {
  subset_data_2 <- DATA_NL %>%
    filter(Experience >= exp_min & Experience <= exp_max)

  result <- lm(log_hourly_wages ~ Education + Afqt, data = subset_data_2)
  return(summary(result))
}

# Run regressions for adjusted experience groups
regression_10_15 <- run_regression_experience(combined_data, 10, 15)
regression_16_21 <- run_regression_experience(combined_data, 16, 21)
regression_21_plus <- run_regression_experience(combined_data, 21, Inf)
```

View regression results

```r
print("Regression for 10-15 Years of Experience")
```

```
## [1] "Regression for 10-15 Years of Experience"
```

```r
print(regression_10_15)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = subset_data_2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.0011 -0.7433 -0.0739  0.6947  5.1663
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.154e-01  1.047e-01   3.012  0.00261 **
## Education   2.348e-01  6.977e-03  33.659  < 2e-16 ***
## Afqt        2.586e-04  4.216e-05   6.134  9.3e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.296 on 4518 degrees of freedom
## Multiple R-squared:  0.2298, Adjusted R-squared:  0.2295
## F-statistic:   674 on 2 and 4518 DF,  p-value: < 2.2e-16
```

```r
print("Regression for 16-21 Years of Experience")
```

```
## [1] "Regression for 16-21 Years of Experience"
```

```r
print(regression_16_21)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = subset_data_2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.2970  -0.7268  -0.0824   0.6660   4.8180
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.8897859  0.1675291  11.280  < 2e-16 ***
## Education   0.1380647  0.0122713  11.251  < 2e-16 ***
## Afqt        0.0002102  0.0000680   3.091  0.00202 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.334 on 1986 degrees of freedom
## Multiple R-squared:  0.07683,    Adjusted R-squared:  0.0759
## F-statistic: 82.64 on 2 and 1986 DF,  p-value: < 2.2e-16
```

```r
print("Regression for 21+ Years of Experience")
```

```
## [1] "Regression for 21+ Years of Experience"
```

```r
print(regression_21_plus)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt, data = subset_data_2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.6880 -0.5769 -0.0466  0.4986  4.4097
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.0802639  0.1347650  22.857   <2e-16 ***
## Education   0.0244069  0.0115114   2.120   0.0346 *
## Afqt        0.0002370  0.0001376   1.722   0.0858 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.106 on 396 degrees of freedom
## Multiple R-squared:  0.02054,    Adjusted R-squared:  0.01559
## F-statistic: 4.151 on 2 and 396 DF,  p-value: 0.01643
```

```r
combined_data$unigrad <- ifelse(combined_data$Education >= 13, 1, 0)
combined_data
```

```
## # A tibble: 19,321 x 15
##    Pubid   Sex  Afqt Annual_hours   Age Education Income  Year weekly_wages
##    <dbl> <dbl> <dbl>        <dbl> <dbl>     <dbl>  <dbl> <int>        <dbl>
##  1     1     2     4         1090    22        16  15000  2003        288.
##  2    31     1   607          370    21        15   3000  2003         57.7
##  3    41     2   405          790    20        14   1000  2003         19.2
##  4    42     1   322          300    20        13   5000  2003         96.2
##  5    49     2   190          526    20        12  15000  2003        288.
##  6    56     2  1024           40    21        15   9000  2003        173.
##  7    62     1   225           60    23        16   3000  2003         57.7
##  8   105     1   305          240    20        12  23000  2003        442.
##  9   111     2   467         1200    22        16   6000  2003        115.
## 10   113     1    55          360    19        12   1200  2003         23.1
## # i 19,311 more rows
## # i 6 more variables: hourly_wages <dbl>, log_hourly_wages <dbl>,
## #   education <dbl>, afqt <dbl>, Experience <dbl>, unigrad <dbl>
```

```r
summary_statistics <- combined_data %>%
  summarise(
    Mean_Log_Wages = mean(log_hourly_wages, na.rm = TRUE),
    SD_Log_Wages = sd(log_hourly_wages, na.rm = TRUE),

    Mean_Education = mean(Education, na.rm = TRUE),
    SD_Education = sd(Education, na.rm = TRUE),

    Mean_AFQT = mean(Afqt, na.rm = TRUE),
    SD_AFQT = sd(Afqt, na.rm = TRUE),

    Mean_unigrad = mean(unigrad, na.rm = TRUE),
    SD_unigrad = sd(unigrad, na.rm = TRUE),
  )

# To view the result
print(summary_statistics)
```

```
## # A tibble: 1 x 8
##   Mean_Log_Wages SD_Log_Wages Mean_Education SD_Education Mean_AFQT SD_AFQT
##            <dbl>        <dbl>          <dbl>        <dbl>     <dbl>   <dbl>
## 1           3.66         1.43           15.1         3.33      590.    477.
## # i 2 more variables: Mean_unigrad <dbl>, SD_unigrad <dbl>
```

```r
summary_statistics <- combined_data %>%
  group_by(Sex) %>%
  summarise(
    Mean_Log_Wages = mean(log_hourly_wages, na.rm = TRUE),
    SD_Log_Wages = sd(log_hourly_wages, na.rm = TRUE),

    Mean_Education = mean(Education, na.rm = TRUE),
    SD_Education = sd(Education, na.rm = TRUE),

    Mean_AFQT = mean(Afqt, na.rm = TRUE),
    SD_AFQT = sd(Afqt, na.rm = TRUE),
```

```r
    Mean_unigrad = mean(unigrad, na.rm = TRUE),
    SD_unigrad = sd(unigrad, na.rm = TRUE),
  )

# To view the result
print(summary_statistics)
```

```
## # A tibble: 2 x 9
##     Sex Mean_Log_Wages SD_Log_Wages Mean_Education SD_Education Mean_AFQT
##   <dbl>          <dbl>        <dbl>          <dbl>        <dbl>     <dbl>
## 1     1           3.73         1.45           14.8         2.95      630.
## 2     2           3.58         1.39           15.6         3.68      542.
## # i 3 more variables: SD_AFQT <dbl>, Mean_unigrad <dbl>, SD_unigrad <dbl>
```

```r
summary_statistics <- combined_data %>%
  group_by(unigrad) %>%
  summarise(
    Mean_Log_Wages = mean(log_hourly_wages, na.rm = TRUE),
    SD_Log_Wages = sd(log_hourly_wages, na.rm = TRUE),

    Mean_Education = mean(Education, na.rm = TRUE),
    SD_Education = sd(Education, na.rm = TRUE),

    Mean_AFQT = mean(Afqt, na.rm = TRUE),
    SD_AFQT = sd(Afqt, na.rm = TRUE),

    Mean_unigrad = mean(unigrad, na.rm = TRUE),
    SD_unigrad = sd(unigrad, na.rm = TRUE),
  )

# To view the result
print(summary_statistics)
```

```
## # A tibble: 2 x 9
##   unigrad Mean_Log_Wages SD_Log_Wages Mean_Education SD_Education Mean_AFQT
##     <dbl>          <dbl>        <dbl>          <dbl>        <dbl>     <dbl>
## 1       0           3.07         1.19           11.3         1.93      408.
## 2       1           3.81         1.44           16.1         2.90      636.
## # i 3 more variables: SD_AFQT <dbl>, Mean_unigrad <dbl>, SD_unigrad <dbl>
```

```r
model_uni <- lm(log_hourly_wages ~ Education + Afqt + unigrad + unigrad:Education + unigrad:Afqt, data =

summary(model_uni )
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt + unigrad +
##     unigrad:Education + unigrad:Afqt, data = combined_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.8073  -0.7845  -0.0458   0.7665   5.8196
```

```
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        2.705e+00  1.320e-01  20.490  < 2e-16 ***
## Education          3.249e-02  1.138e-02   2.854 0.004321 **
## Afqt              -6.436e-06  5.833e-05  -0.110 0.912143
## unigrad           -5.204e-01  1.461e-01  -3.562 0.000369 ***
## Education:unigrad  5.952e-02  1.201e-02   4.955  7.3e-07 ***
## Afqt:unigrad       2.265e-04  6.261e-05   3.618 0.000297 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.368 on 19315 degrees of freedom
## Multiple R-squared:  0.07874,    Adjusted R-squared:  0.0785
## F-statistic: 330.2 on 5 and 19315 DF,  p-value: < 2.2e-16
```

```r
combined_data_exp <- combined_data[combined_data$Experience > 15, ]
model_uni_exp <- lm(log_hourly_wages ~ Education + Afqt + unigrad + unigrad:Education + unigrad:Afqt, da
summary(model_uni_exp)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt + unigrad +
##     unigrad:Education + unigrad:Afqt, data = combined_data_exp)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.2631  -0.6845  -0.0806   0.6183   4.7808
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        3.003e+00  1.419e-01  21.155  < 2e-16 ***
## Education          4.655e-02  1.245e-02   3.739 0.000189 ***
## Afqt              -5.313e-05  1.139e-04  -0.467 0.640808
## unigrad           -1.348e+00  3.842e-01  -3.508 0.000461 ***
## Education:unigrad  1.030e-01  2.702e-02   3.812 0.000142 ***
## Afqt:unigrad       3.718e-04  1.374e-04   2.707 0.006846 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.31 on 2227 degrees of freedom
## Multiple R-squared:  0.08641,    Adjusted R-squared:  0.08436
## F-statistic: 42.13 on 5 and 2227 DF,  p-value: < 2.2e-16
```

```r
combined_data_exp2 <- combined_data[combined_data$Experience < 15, ]
model_uni_exp2 <- lm(log_hourly_wages ~ Education + Afqt + unigrad + unigrad:Education + unigrad:Afqt, d
summary(model_uni_exp2)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt + unigrad +
##     unigrad:Education + unigrad:Afqt, data = combined_data_exp2)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.8904  -0.7773  -0.0438   0.7729   5.8799
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        1.339e+00  3.275e-01   4.088 4.38e-05 ***
## Education          1.391e-01  2.811e-02   4.949 7.53e-07 ***
## Afqt              -2.421e-06  6.813e-05  -0.036   0.9717
## unigrad            7.627e-01  3.337e-01   2.285   0.0223 *
## Education:unigrad -4.527e-02  2.838e-02  -1.595   0.1107
## Afqt:unigrad       2.111e-04  7.227e-05   2.920   0.0035 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.364 on 16469 degrees of freedom
## Multiple R-squared:  0.08731,    Adjusted R-squared:  0.08703
## F-statistic: 315.1 on 5 and 16469 DF,  p-value: < 2.2e-16
```

```r
# for men only
combined_data_men <- combined_data[combined_data$Sex == 1, ]
model_uni_men <- lm(log_hourly_wages ~ Education + Afqt + unigrad + unigrad:Education + unigrad:Afqt, da
summary(model_uni_men)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt + unigrad +
##     unigrad:Education + unigrad:Afqt, data = combined_data_men)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.6573  -0.8037  -0.0784   0.7516   5.7106
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        2.165e+00  2.195e-01   9.866  < 2e-16 ***
## Education          9.344e-02  1.886e-02   4.955 7.35e-07 ***
## Afqt              -1.224e-04  7.240e-05  -1.691   0.0909 .
## unigrad           -2.398e-01  2.432e-01  -0.986   0.3243
## Education:unigrad  2.057e-02  2.000e-02   1.029   0.3036
## Afqt:unigrad       3.747e-04  7.875e-05   4.758 1.98e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.384 on 10413 degrees of freedom
## Multiple R-squared:  0.08991,    Adjusted R-squared:  0.08947
## F-statistic: 205.7 on 5 and 10413 DF,  p-value: < 2.2e-16
```

Education (Education): The coefficient is positive and significant ($p < 0.05$), suggesting that higher levels of education are associated with higher log wages for men. This is consistent with the expectation that education increases productivity, which in turn is rewarded in the labor market.

Cognitive Ability (AFQT): The coefficient is negative and not statistically significant ($p > 0.05$), which is not in line with the typical expectation that higher cognitive abilities are associated with higher wages. The

lack of significance suggests that, for men in this sample, AFQT scores may not be a strong predictor of wage levels.

University Graduate (unigrad): The coefficient is negative but not statistically significant, indicating that simply being a university graduate may not have a distinct impact on wages separate from the number of years of education completed.

Interaction between University Graduate and Education (unigrad:Education): The coefficient is negative and significant, which could imply that additional years of education beyond achieving a university degree may have diminishing returns in terms of wages for men, or that the effect of education on wages is less pronounced for university graduates compared to those with less education.

Interaction between University Graduate and AFQT (unigrad:AFQT): The coefficient is positive and significant, suggesting that for university graduates, higher AFQT scores may be more strongly correlated with wages compared to non-graduates.

```r
# for women only
combined_data_women <- combined_data[combined_data$Sex == 2, ]
model_uni_women <- lm(log_hourly_wages ~ Education + Afqt + unigrad + unigrad:Education + unigrad:Afqt,
summary(model_uni_women)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt + unigrad +
##     unigrad:Education + unigrad:Afqt, data = combined_data_women)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.5959 -0.7581 -0.0050  0.8003  5.0688
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        2.934e+00  1.625e-01  18.051  < 2e-16 ***
## Education         -1.764e-02  1.410e-02  -1.251 0.211078
## Afqt               2.077e-04  9.826e-05   2.113 0.034586 *
## unigrad           -6.671e-01  1.801e-01  -3.705 0.000212 ***
## Education:unigrad  1.028e-01  1.484e-02   6.924 4.71e-12 ***
## Afqt:unigrad      -1.346e-04  1.041e-04  -1.293 0.196007
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.333 on 8896 degrees of freedom
## Multiple R-squared:  0.0832, Adjusted R-squared:  0.08269
## F-statistic: 161.5 on 5 and 8896 DF,  p-value: < 2.2e-16
```

```r
combined_data$female <- ifelse(combined_data$Sex == 2, 1, 0)
```

```r
model_fem <- lm(log_hourly_wages ~ Education + Afqt + female + female:Education + female:Afqt, data = c

summary(model_fem)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt + female + female:Education +
```

```
##     female:Afqt, data = combined_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.9152  -0.7970  -0.0499   0.7862   5.7243
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)       1.677e+00  6.832e-02  24.553  < 2e-16 ***
## Education         1.302e-01  4.747e-03  27.416  < 2e-16 ***
## Afqt              2.053e-04  2.799e-05   7.337 2.28e-13 ***
## female            3.466e-01  9.336e-02   3.713 0.000206 ***
## Education:female -3.447e-02  6.196e-03  -5.563 2.68e-08 ***
## Afqt:female      -9.292e-05  4.336e-05  -2.143 0.032127 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.367 on 19315 degrees of freedom
## Multiple R-squared:  0.08087,    Adjusted R-squared:  0.08063
## F-statistic: 339.9 on 5 and 19315 DF,  p-value: < 2.2e-16
```

Education: The coefficient for education is significant and positive, indicating that an additional year of education is associated with an increase in the log of hourly wages.

AFQT: The coefficient for AFQT scores is also significant and positive, suggesting that higher AFQT scores are associated with higher log hourly wages.

Female: The 'female' coefficient is significant and positive, implying that being female is associated with an increase in log hourly wages, holding other factors constant.

Education:female: The interaction term between education and female is significant and negative. This suggests that the positive impact of education on log hourly wages is less for females than for males.

Afqt:female: The interaction term between AFQT scores and female is significant but has a negative coefficient, indicating that the positive impact of AFQT scores on log hourly wages is less for females compared to males.

```r
clustered_covariance <- vcovHC(model_fem, type = "HC1", cluster = "group", group = combined_data$Pubid)

clustered_summary <- coeftest(model_fem, vcov = clustered_covariance)

print(clustered_summary)
```

```
##
## t test of coefficients:
##
##                     Estimate  Std. Error t value  Pr(>|t|)
## (Intercept)       1.6774e+00  2.8997e-01  5.7846 7.379e-09 ***
## Education         1.3015e-01  2.1264e-02  6.1207 9.499e-10 ***
## Afqt              2.0533e-04  4.4682e-05  4.5953 4.349e-06 ***
## female            3.4662e-01  3.5753e-01  0.9695    0.3323
## Education:female -3.4471e-02  2.5417e-02 -1.3562    0.1750
## Afqt:female      -9.2924e-05  5.6665e-05 -1.6399    0.1010
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
combined_data$lowexp <- ifelse(combined_data$Experience <= 5, 1, 0)

pooled_regression_model <- lm(log_hourly_wages ~ Education + Afqt + lowexp + lowexp:Education + lowexp:

summary(pooled_regression_model)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt + lowexp + lowexp:Education +
##     lowexp:Afqt, data = combined_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.0878  -0.7674  -0.0788   0.7489   6.0487
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)       1.025e+00  6.272e-02  16.336   <2e-16 ***
## Education         1.836e-01  4.324e-03  42.453   <2e-16 ***
## Afqt              2.614e-04  2.755e-05   9.487   <2e-16 ***
## lowexp            1.242e+00  9.070e-02  13.693   <2e-16 ***
## Education:lowexp -1.202e-01  5.913e-03 -20.336   <2e-16 ***
## Afqt:lowexp      -1.037e-04  4.110e-05  -2.523   0.0117 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.32 on 19315 degrees of freedom
## Multiple R-squared:  0.1422, Adjusted R-squared:  0.1419
## F-statistic: 640.1 on 5 and 19315 DF,  p-value: < 2.2e-16
```

Education is a highly significant predictor of log hourly wages, indicating that as the education level increases, the log hourly wages tend to increase. The AFQT score is also significant, showing that higher cognitive abilities, as measured by the AFQT, are associated with higher log hourly wages. The variable lowexp (low experience, assumedly for those with 5 years or less of experience) is significant, suggesting that less experience is linked with lower wages, but this effect is mitigated by education and AFQT scores, as indicated by the interaction terms. The interaction between low experience and education (Education:lowexp) is negatively significant, which could mean that the positive effect of education on wages is less pronounced for those with lower experience. The interaction between low experience and AFQT scores (Afqt:lowexp) is significant, indicating that the relationship between cognitive ability and wages is different for those with low experience compared to those with more experience.

```r
# Separate data by gender for additional regressions
# Assuming that 'Sex' is coded as 1 for males and 2 for females
data_men <- combined_data %>% filter(Sex == 1)
data_women <- combined_data %>% filter(Sex == 2)

# Run the regression for men
men_regression_model <- lm(log_hourly_wages ~ Education + Afqt + lowexp + lowexp:Education + lowexp:Afq

# Summary of the men's regression
summary(men_regression_model)
```

```
##
```

```
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt + lowexp + lowexp:Education +
##     lowexp:Afqt, data = data_men)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.2353  -0.7775  -0.1057   0.7115   6.0319
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)       7.396e-01  8.596e-02   8.603  < 2e-16 ***
## Education         2.159e-01  6.211e-03  34.768  < 2e-16 ***
## Afqt              2.053e-04  3.620e-05   5.672 1.45e-08 ***
## lowexp            1.391e+00  1.384e-01  10.048  < 2e-16 ***
## Education:lowexp -1.455e-01  9.406e-03 -15.466  < 2e-16 ***
## Afqt:lowexp       2.911e-05  5.459e-05   0.533    0.594
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.321 on 10413 degrees of freedom
## Multiple R-squared:  0.1711, Adjusted R-squared:  0.1707
## F-statistic: 429.8 on 5 and 10413 DF,  p-value: < 2.2e-16
```

All the predictors, including the interaction terms, are significant for men, except for the interaction between low experience and AFQT scores ($p > 0.05$), which is not significant. This suggests that while education and low experience have a strong relationship with log hourly wages, the impact of AFQT scores doesn't differ significantly between men with less or more experience.

```r
# Run the regression for women
women_regression_model <- lm(log_hourly_wages ~ Education + Afqt + lowexp + lowexp:Education + lowexp:Af

# Summary of the women's regression
summary(women_regression_model)
```

```
##
## Call:
## lm(formula = log_hourly_wages ~ Education + Afqt + lowexp + lowexp:Education +
##     lowexp:Afqt, data = data_women)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.2770 -0.7529 -0.0446  0.7781  4.4961
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)       1.063e+00  9.337e-02  11.385  < 2e-16 ***
## Education         1.711e-01  6.142e-03  27.856  < 2e-16 ***
## Afqt              1.934e-04  4.321e-05   4.476 7.72e-06 ***
## lowexp            1.300e+00  1.251e-01  10.386  < 2e-16 ***
## Education:lowexp -1.106e-01  7.851e-03 -14.090  < 2e-16 ***
## Afqt:lowexp      -1.693e-04  6.365e-05  -2.660  0.00782 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 1.304 on 8896 degrees of freedom
## Multiple R-squared:  0.1229, Adjusted R-squared:  0.1224
## F-statistic: 249.4 on 5 and 8896 DF,  p-value: < 2.2e-16
```

For women, the predictors are similar to the pooled model, with significant effects of education, AFQT, and low experience on log hourly wages. The interaction between low experience and education (Education:lowexp) for women is significantly negative, similar to the pooled regression, suggesting that the positive effect of education on wages is less for those with low experience. The interaction between low experience and AFQT (Afqt:lowexp) for women is also negatively significant, indicating that the effect of cognitive abilities on wages is also less for women with low experience.